

HP PCIe IO Accelerator for ProLiant Servers Installation Guide

Abstract

Use these instructions to install the HP PCIe IO Accelerator device.



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Contents

- Overview 4
 - Software compatibility 4
 - In the box 4
- Hardware specifications and requirements 5
 - Hardware requirements 5
- Installation 9
 - Warnings and cautions 9
 - Removing the access panel 9
 - Installing the HP PCIe IO Accelerator device 9
 - Installing a half-height bracket 13
- Regulatory compliance notices 16
 - Safety and regulatory compliance 16
- Acronyms and abbreviations 17
- Documentation feedback 18

Overview

Software compatibility

Compatible software (driver)

Each release of the HP IO Accelerator VSL software is compatible with certain IO Accelerator devices.

For a list of devices that are compatible with the version of the IO Accelerator VSL software you are installing, see the *HP IO Accelerator Driver and Management Software Release Notes* for that version.

Compatible operating systems

The operating system requirements depend on the version of IO Accelerator VSL software that you are installing with the device.

For more information, see the *HP IO Accelerator Driver and Management Software Release Notes* for that version.

In the box

The HP IO Accelerator device comes with the following items:

- HP IO Accelerator device
- Installation instructions
- Warranty

Depending on the device, the following additional item might be included:

- Half-height bracket, used on low-profile systems or systems with low-profile PCIe slots

Hardware specifications and requirements

Hardware requirements

Before installing the IO Accelerator device, be sure your system hardware meets the following requirements.

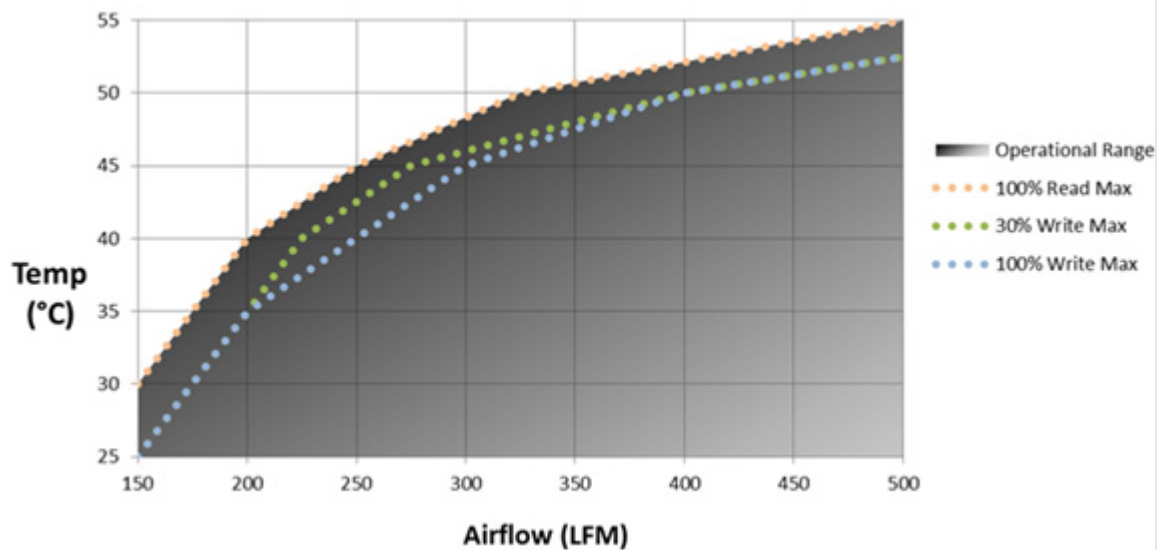
Adequate system cooling

Provide adequate system cooling to maximize the longevity and performance of IO Accelerator devices. System cooling is provided by both the ambient temperature as well as airflow across the devices.

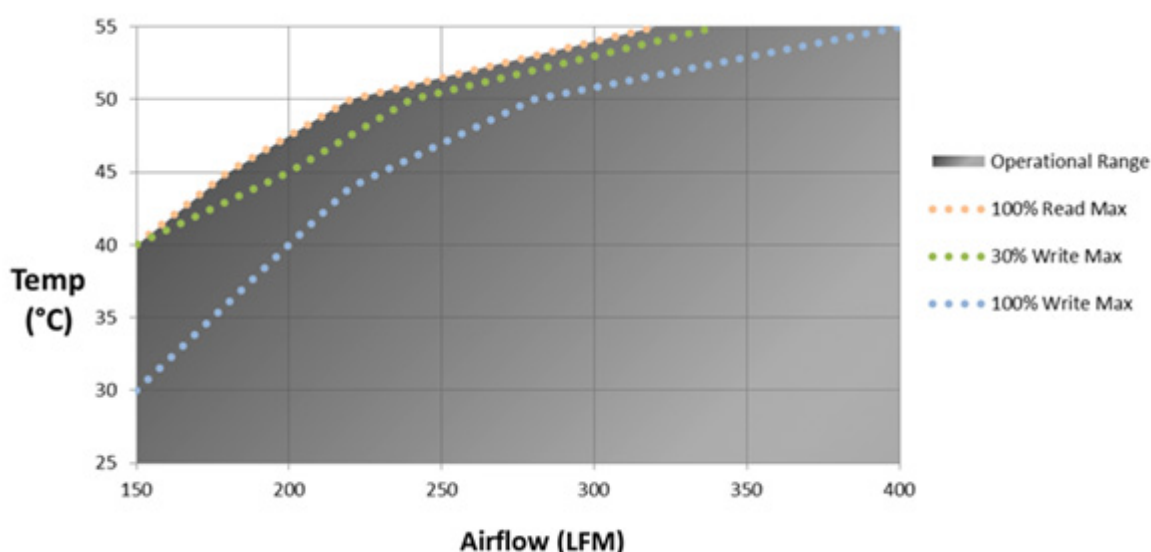
The following graphs show the amount of cooling required for the IO Accelerator device depending on the physical design of the device (low profile or standard, half-length devices).

The graphs are based on tests performed in a ducted (3.13" x 3.50" cross-section) wind tunnel with generally laminar air flow (three units placed side-by-side in a ducted section, with data recorded from the center unit). Airflow and temperature profiles will vary in specific host environments. These graphs represent worst-case conditions.

The following graph shows the cooling required for a low profile IO Accelerator device where the temperature is the ambient temperature of the air, in °C, as it comes in contact with the device. For example, if your inlet temperature to the device was 48°C, you need airflow of at least 300 LFM in order to operate the device at 100% reads.



The following graph shows the cooling required for a standard height, half length IO Accelerator device where the temperature is the ambient temperature of the air, in °C, as it comes in contact with the device. For example, if your inlet temperature to the device was 48°C, you need airflow of at least 220 LFM in order to operate the device at 100% reads.



Device thermal monitoring

In order to protect against thermal damage, all IO Accelerator devices monitor the temperature of the onboard controller chip (the temperature is reported by the `fio-status` command-line utility as `Internal temperature`) as well as the NAND flash to prevent overheating of the NAND flash components (reported by `fio-status`, using the `-fj` or `-fx` options, as `nand_thermal_deg_c`). If your device is throttled or shuts down due to thermal issues, the `fio-status` utility prints a warning and indicates whether the cause was the `Controller Junction` or the `NAND Temperature`.

- In an attempt to contain temperatures within an optimal range, the IO Accelerator VSL software starts throttling write performance when the controller temperature or NAND reaches a set temperature.
- If the controller or NAND temperature continues to rise, the software powers down the device when the maximum operating temperature is reached.

The throttling and shutdown temperature might depend on the device.

To find the temperature limits for your device, find your device in the section "Thermal thresholds and PCIe slot requirements."

If your system logs indicate write-performance throttling due to high temperatures, consult your server documentation for details on increasing airflow within your system (for example, by increasing the fans to maximum speed within the system via a BIOS setting).



IMPORTANT: If available, enable the BIOS has a High Performance/Power Mode when using IO Accelerator devices. Disable any power-saving modes. The settings improve performance in two ways:

- The settings prevent the operating system and/or the BIOS from using ASPM to suspend PCIe devices, including IO Accelerator devices. IO Accelerator devices do not support ASPM.
- The settings maintain higher fan speeds to prevent thermal throttling.

Thermal thresholds and PCIe slot requirements

The minimum slot requirements for your IO Accelerator device depends on the type of product.

After you install the IO Accelerator VSL software, use the `fio-pci-check` utility on most operating systems to see if the PCIe slot meets the bandwidth requirements.

Devices	Controller throttle temperature	Controller shutdown temperature	NAND board throttle temperature	NAND board shutdown temperature
SX300 devices	78°C	85°C	78°C	83°C
PX600 devices	78°C	85°C	78°C	83°C

Devices	Minimum PCIe slot	Slot clearance
SX300-1300 SX300-1305 SX300-1600 SX300-1605 SX300-3200 SX300-3205	x8 slot*	Low profile
SX300-6400 SX300-6405	x8 slot*	Standard height, half length
SX300-1200 Mezzanine SX300-1600 Mezzanine	-	Type B mezzanine
PX600-1000 PX600-1005 PX600-1300 PX600-1305 PX600-2600 PX600-2605	x8 slot*	Low profile
PX600-5200 PX600-5205	x8 slot*	Standard height, half length

*An x8 slot is the minimum physical requirement. A Gen2 x8 slot is required for full performance.

Sufficient RAM

The IO Accelerator VSL software requires enough RAM to accelerate the IO Accelerator device. The RAM requirements depend on the capacity of your IO Accelerator devices, the version of the IO Accelerator VSL software in use on the system, and the way in which the operating system/file system tracks I/O (for example, average written block size).

For more information, including a chart of RAM required per GB of IO Accelerator device capacity, see the *HP IO Accelerator Driver and Management Software Release Notes* for the version of IO Accelerator VSL software that you install with the devices.

Firmware requirements

The IO Accelerator device must have a certain level of firmware in order to work with particular versions of the IO Accelerator VSL software. For upgrade considerations, see the "Upgrade notes" section of the *HP IO Accelerator Driver and Management Software Release Notes*.



CAUTION: Do not downgrade the IO Accelerator device to an earlier version of the firmware. Earlier versions of the firmware might not be compatible with the device, and downgrading the firmware might result in data loss. If you must downgrade your firmware, contact HP Customer Support (<http://www.hp.com/support>) for compatibility information and to discuss your use case.

Installation

Warnings and cautions

-
-  **WARNING:** To reduce the risk of personal injury or damage to the equipment, consult the safety information and user documentation provided with the server before attempting the installation. Some servers contain high energy circuits, high current circuits, moving parts (such as fan blades), or any combination of these hazards, that may be exposed if covers and access panels are removed while the product is connected to a power source. These products are intended to be serviced only by qualified personnel who have been trained to deal with these hazards. Do not remove enclosures or attempt to bypass any interlocks designed to guard against these hazardous conditions.
-
-  **WARNING:** To reduce the risk of electric shock, remove all AC power from the system before installing or replacing any non-hot-plug hardware option to the system. Disconnect all power cords to completely remove power from the server.
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-  **WARNING:** To reduce the risk of personal injury from hot surfaces, allow the drives and the internal system components to cool before touching them.
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Removing the access panel

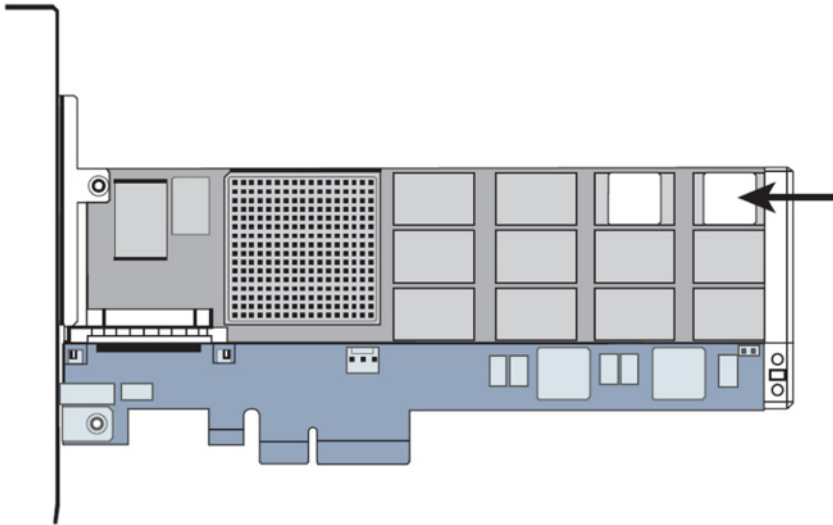
1. Power down the server.
2. Disconnect the power cord from the server.
3. Remove the server access panel.

For detailed instructions on powering down the server and removing the access panel, see the server user guide.

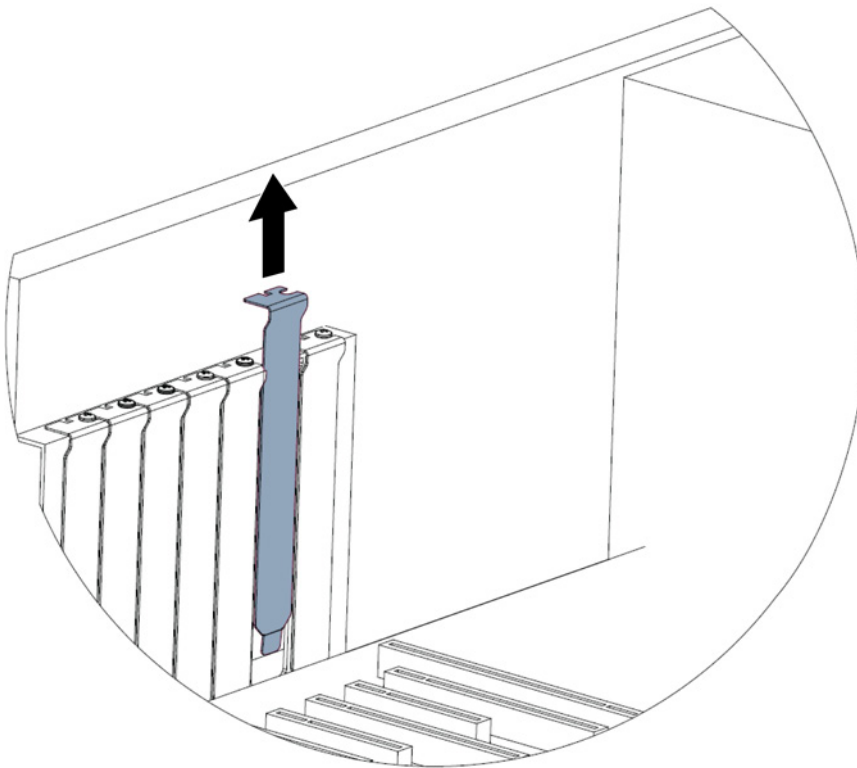
Installing the HP PCIe IO Accelerator device

1. If you are installing the HP PCIe IO Accelerator device in a low-profile system, install the half-height bracket before installing the PCIe IO Accelerator device. For more information, see "Installing a half-height bracket (on page 13)."
2. Locate and record the serial number on your HP IO Accelerator device. The serial number label has a number and a barcode.

After the IO Accelerator device is installed, the device serial number is visible in the `fio-status` IO Accelerator VSL software utility.



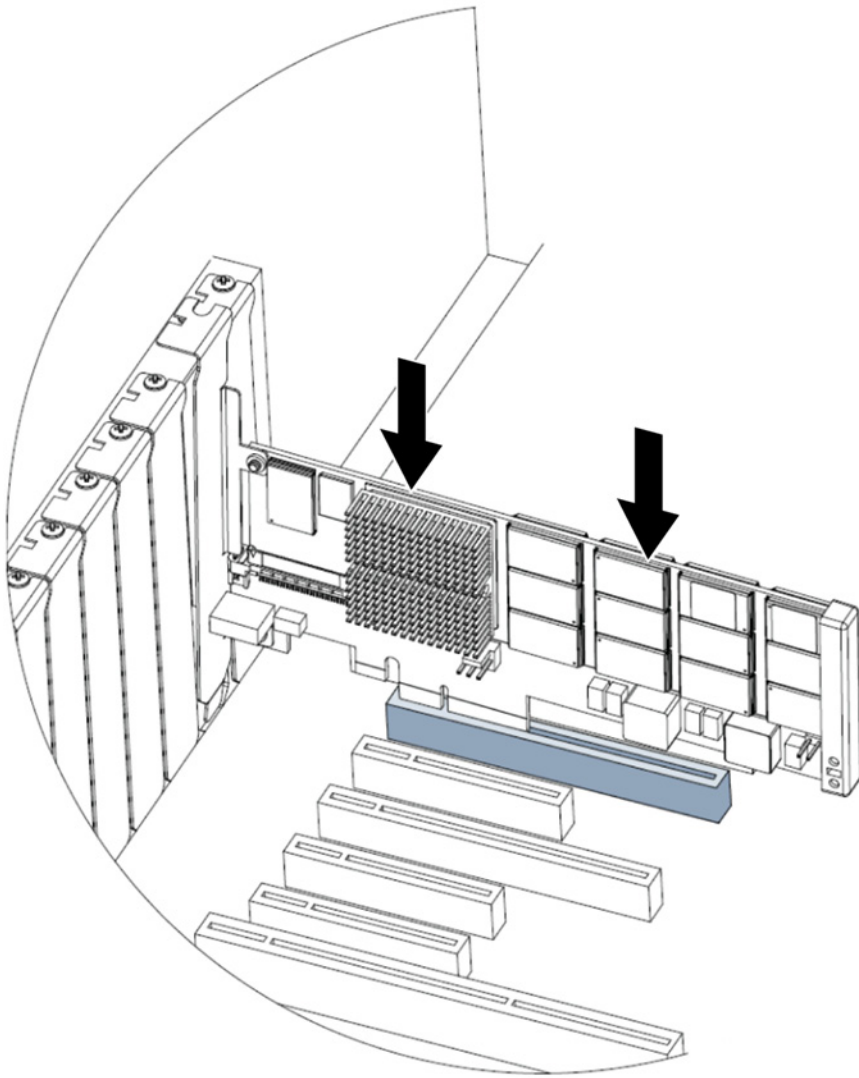
3. Locate an empty PCIe slot compatible with the device. For details on identifying PCIe slots, see your server documentation.
4. If present, remove the slot bracket.



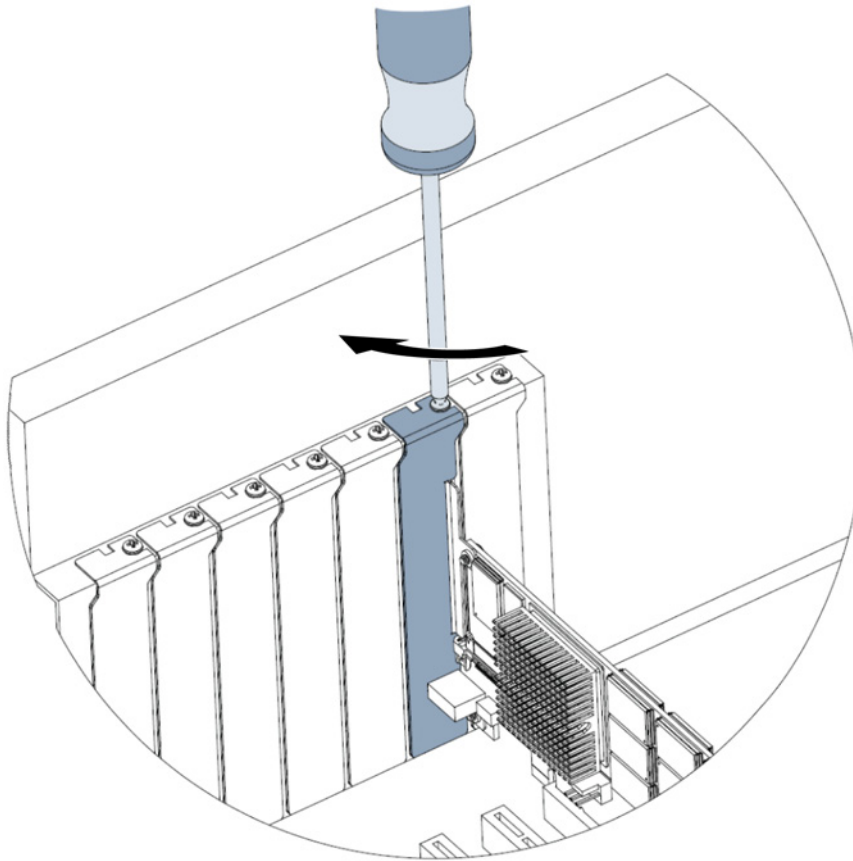
5. Grasp the IO Accelerator device by the top edge and insert it gently, but firmly, into the available PCIe slot.



CAUTION: To prevent damage to the server, apply pressure over the IO Accelerator connector when installing the IO Accelerator device. Do not apply pressure to the edges of the IO Accelerator device.



6. Depending on how your hardware is configured, use a screw or lever to secure the IO Accelerator retaining bracket.



7. Replace the server access panel.
8. Connect the AC power cord to the server.
9. Power up the server.
10. If your operating system detects the HP IO Accelerator device and prompts you to install a driver for the device, click **Cancel**.

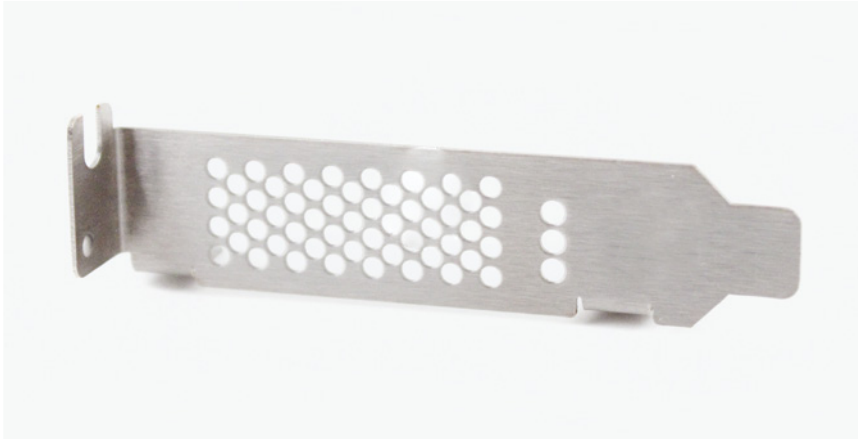
The installation is complete.

You are now ready to install the drivers and utilities software. For more information, see the *HP IO Accelerator User Guide*.

Installing a half-height bracket

For half-height installation (such as in low-profile systems), replace the full-height retaining bracket with the half-height bracket included in your HP IO Accelerator device kit:

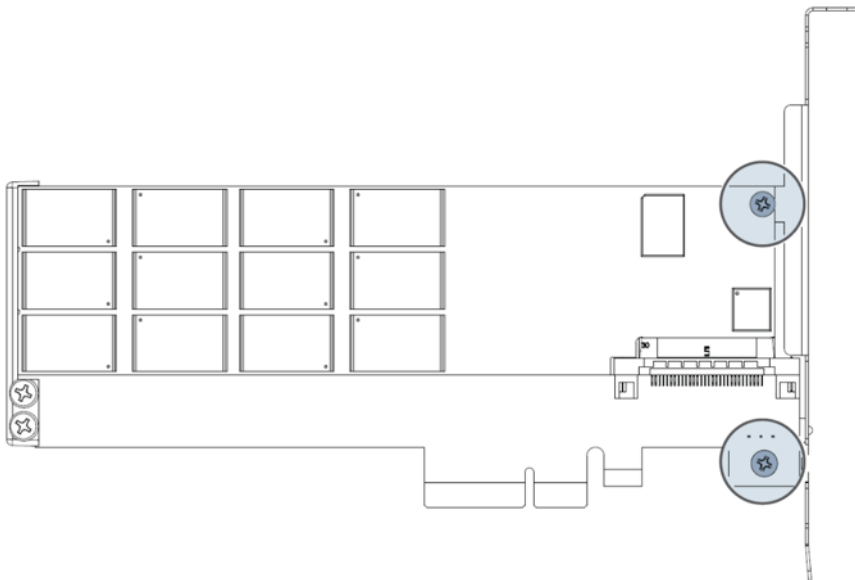
1. Locate the half-height bracket in your IO Accelerator device package.



CAUTION: Use care when removing the retaining screws. Do not twist or pull on the bracket until both screws are out as this can cause damage to components.

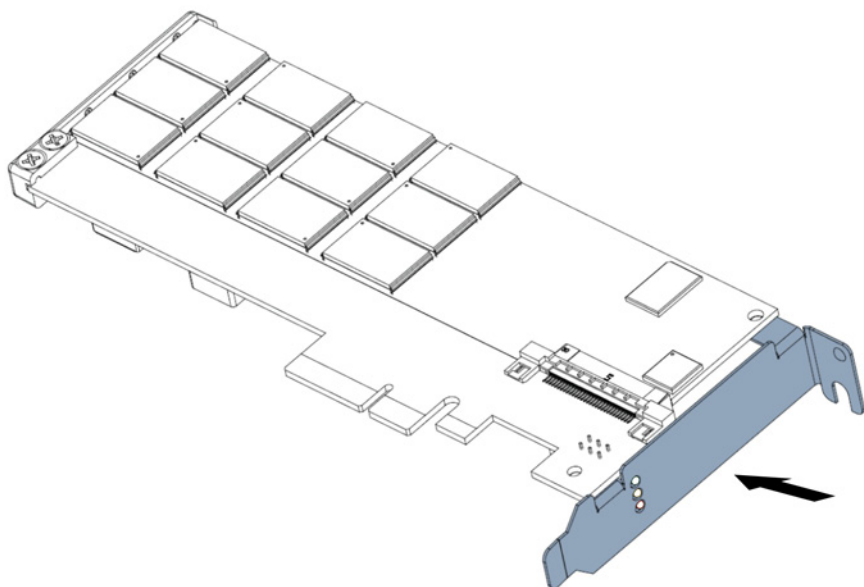
To prevent damage to the PCIe IO Accelerator, use only a No. 1 Philips screwdriver.

2. Remove the two screws holding the full-height bracket to the IO Accelerator device.

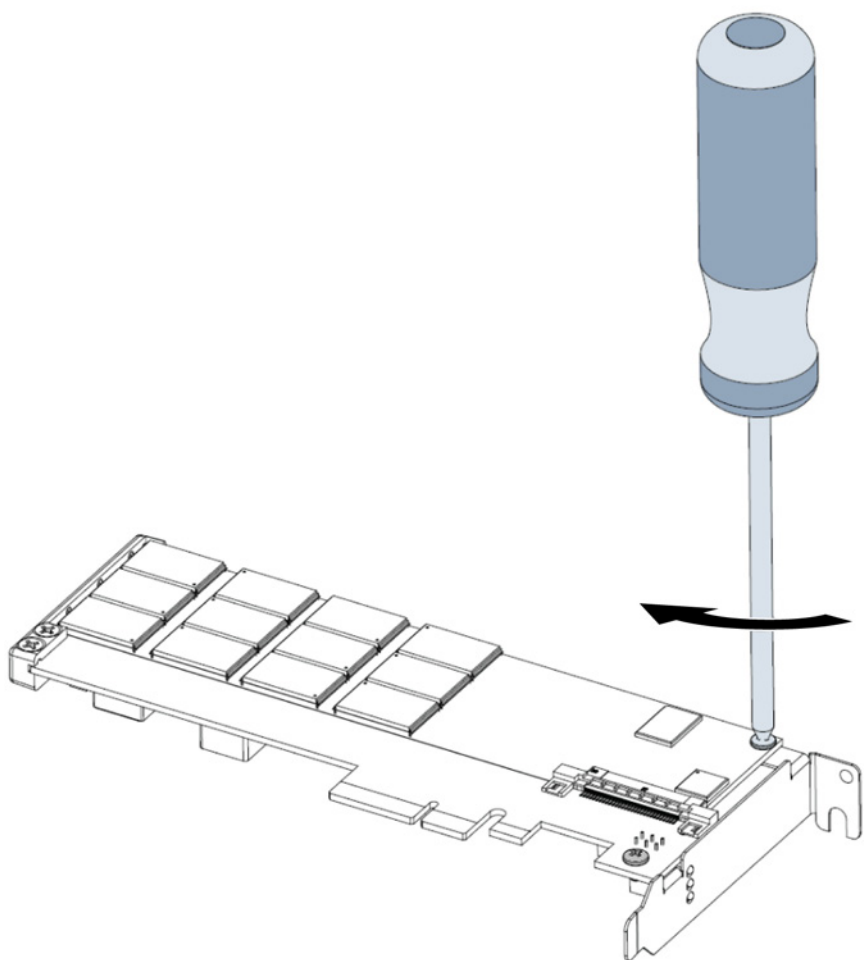


3. Remove the bracket carefully from the device.

4. Align the LEDs on the IO Accelerator device with the holes in the half-height bracket. Be sure the bracket tabs are on the heatsink side of the device.



5. Attach the half-height bracket by using a No. 1 Philips screwdriver to tighten the two screws.





CAUTION: Do not overly tighten screws. Overly tightening the screws might damage the device.

6. Return to "Installing the HP PCIe IO Accelerator device (on page 9)," and begin with step 2.

Regulatory compliance notices

Safety and regulatory compliance

For safety, environmental, and regulatory information, see *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at the HP website (<http://www.hp.com/support/Safety-Compliance-EnterpriseProducts>).

Acronyms and abbreviations

ASPM

active state power management

GPU

graphics processing unit

PCIe

Peripheral Component Interconnect Express

Documentation feedback

HP is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (<mailto:docsfeedback@hp.com>). Include the document title and part number, version number, or the URL when submitting your feedback.